

End-to-End & Integration Test Cases

Project: FlakyTest Detector Engine

Team: Team Syndicate

This document outlines the comprehensive manual, integration, and user acceptance test cases (UAT) executed to validate the end-to-end functionality of the FlakyTest Detector Engine.

1. File Upload & Data Ingestion

Test ID	Scenario	Expected Result	Status
TC-01	Upload valid standard CSV file	File uploads successfully. UI displays success toast. Backend parses all rows.	PASS
TC-02	Upload unformatted JSON file	Universal Parser infers schema and normalizes data.	PASS
TC-03	Upload corrupted/empty file	System rejects file and displays clear error message.	PASS
TC-04	Missing mandatory columns	Parser applies defaults and logs warnings without failure.	PASS

2. Flaky Test Detection Engine

Test ID	Scenario	Expected Result	Status
TC-05	Stable test filtering	Stable tests are excluded from flaky list.	PASS
TC-06	Flakiness Score Calculation	Correct flakiness percentage displayed.	PASS
TC-07	Severity Color Coding	LOW, MEDIUM, HIGH severity levels displayed correctly.	PASS

3. AI Agent & Root-Cause Analysis

Test ID	Scenario	Expected Result	Status
TC-08	Generate AI Explanation (Local)	Local Ollama returns structured analysis.	PASS
TC-09	LLM Fallback Mechanism	System falls back to Groq API when Ollama is unavailable.	PASS
TC-10	Chat Interface Context	Agent answers using dataset context.	PASS
TC-11	Prompt Injection Prevention	Off-topic prompts are rejected.	PASS

4. UI/UX & Responsive Design

Test ID	Scenario	Expected Result	Status
TC-12	Mobile Responsiveness	Responsive layout adapts correctly.	PASS
TC-13	Dark Mode Consistency	UI theme remains visually consistent.	PASS
TC-14	Loading States	Spinners and disabled actions prevent duplicate requests.	PASS

5. API Endpoints & Security

Test ID	Scenario	Expected Result	Status
TC-15	CORS Policy Validation	Unauthorized requests are blocked.	PASS
TC-16	API Rate Limiting / Timeout	Slow requests timeout safely.	PASS

Test Execution Summary

Total Test Cases: 16

Passed: 16

Failed: 0

Pass Rate: 100%

The system successfully passed all end-to-end, integration, UI, AI, and security validation scenarios.